Digital Logic - CISC221

— Assignment 4 —

Arham Hasson
—20426387—
23cr8@queensu.ca

Q1) Boolean Expression in sum-of-product form, minterms in ascending binary order. (and the rest)

(a) So, $Y = \bar{A}\bar{B} + A\bar{B} + AB$ //

$m_0 \rightarrow m_2 \rightarrow m_3$

→ Sigma notation = $\Sigma(0,2,3)$ //

→ K-maps simplification:

	A	B	Y
m_0	0	0	1 = $\bar{A}\bar{B}$
m_1	0	1	0
m_2	1	0	1 = $A\bar{B}$
m_3	1	1	1 = AB

A \ B	0	1
0	m_0	m_1
1	m_2	m_3

→

A \ B	0	1
0	1	0
1	1	1

→ 00, 01 = $\bar{B}$, 10, 11 = A } $\bar{B} + A$ //

→ Simplified expression: $Y = \bar{B} + A$

Part (b)

~~So, $Y = \bar{A}+\bar{B}+\bar{C}+A$~~

→ So, $Y = \bar{A}\bar{B}\bar{C} + ABC$ //

→ Sigma Notation = $\Sigma(0,7)$ //

→ K-maps:

	A	B	C	Y
m_0	0	0	0	1 = $\bar{A}+\bar{B}+\bar{C}$
m_1	0	0	1	0
m_2	0	1	0	0
m_3	0	1	1	0
m_4	1	0	0	0
m_5	1	0	1	0
m_6	1	1	0	0
m_7	1	1	1	1 = $A+B+C$

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

A \ BC	00	01	11	10
0	1	0	0	0
1	0	0	1	0

→ gives 000 and 111, no adjacent groups so = find ⇒ $Y = \bar{A}\bar{B}\bar{C} + ABC$ //

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Part (c)

Part (d)

1) $Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + ABC$ //

	A	B	C	Y	
$\bar{A}\bar{B}\bar{C}$ =	0	0	0	1	m_0
	0	0	1	0	m_1
$\bar{A}B\bar{C}$ =	0	1	0	1	m_2
	0	1	1	0	m_3
$A\bar{B}\bar{C}$ =	1	0	0	1	m_4
$A\bar{B}C$ =	1	0	1	1	m_5
	1	1	0	0	m_6
ABC =	1	1	1	1	m_7

2) Sigma Notation: $\Sigma(0,2,4,5,7)$ //

"I forgot to show the steps of getting the Sigma notation but I just did → $\bar{A}\bar{B}\bar{C} = 000$, $\bar{A}B\bar{C} = 010$, $A\bar{B}\bar{C} = 100$, $A\bar{B}C = 101$, $ABC = 111$ → converting binary to decimal

Binary =	000	010	100	101	111
Dec..	0	2	4	5	7

(I skipped this step in (a) and (b) as I thought it's a fairly straightforward calculation, but here's how I did those anyways explained)

3) K-maps and Simplification:

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

A \ BC	00	01	11	10
0	1	0	0	1
1	1	1	1	0

Group 1 = m_0/m_2
" 2 = m_4/m_6
" 3 = m_5/m_7

$= \bar{A}\bar{C} + A\bar{B} + AC$
$= 00 + 10 + 11$

→ Y (simplified) $= \bar{A}\bar{C} + A\bar{B} + AC$
$= A(\bar{B} + C) + \bar{A}\bar{C}$

So our final answer is

$$Y = A(\bar{B} + C) + \bar{A}\bar{C} \quad //$$

Part (d)

	A	B	C	D	Y	
$\bar{A}\bar{B}\bar{C}\bar{D}$ =	0	0	0	0	1	m_0
$\bar{A}\bar{B}\bar{C}D$ =	0	0	0	1	1	m_1
$\bar{A}\bar{B}C\bar{D}$ =	0	0	1	0	1	m_2
$\bar{A}\bar{B}CD$ =	0	0	1	1	1	m_3
	0	1	0	0	0	m_4
	0	1	0	1	0	m_5
	0	1	1	0	0	m_6
	0	1	1	1	0	m_7
$A\bar{B}\bar{C}\bar{D}$ =	1	0	0	0	1	m_8
	1	0	0	1	0	m_9
$A\bar{B}C\bar{D}$ =	1	0	1	0	1	m_{10}
	1	0	1	1	0	m_{11}
	1	1	0	0	0	m_{12}
	1	1	0	1	0	m_{13}
$ABC\bar{D}$ =	1	1	1	0	1	m_{14}
	1	1	1	1	0	m_{15}

1) $Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + ABC\bar{D}$ //

2) Sigma Notation → 0000, 0001, 0010, 0011, 1000, 1010 + 1110

$= \Sigma(0, 1, 2, 3, 8, 10, 14)$ //

3) K-Maps (simplification):

AB \ CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m8	m9	m11	m10
10	m12	m13	m15	m14

→

3) K-maps (simplified):

AB \ CD	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	1
10	1	0	0	1

AB \ CD	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

→ 3 Groups = $\bar{A}\bar{B}$, $\bar{B}\bar{D}$, $AC\bar{D}$

So, Simplified Expression:

$$Y = \bar{A}\bar{B} + \bar{B}\bar{D} + AC\bar{D}$$

Part (e)

1) $Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + ABCD$

2) Sigma notation:

0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111

$= \Sigma(0,3,5,6,9,10,12,15)$

	A	B	C	D	Y	
$\bar{A}\bar{B}\bar{C}\bar{D}$ =	0	0	0	0	1	m_0
	0	0	0	1	0	m_1
	0	0	1	0	0	m_2
$\bar{A}\bar{B}CD$ =	0	0	1	1	1	m_3
	0	1	0	0	0	m_4
$\bar{A}B\bar{C}\bar{D}$ =	0	1	0	1	1	m_5
$\bar{A}BC\bar{D}$ =	0	1	1	0	1	m_6
	0	1	1	1	0	m_7
	1	0	0	0	0	m_8
$A\bar{B}\bar{C}D$ =	1	0	0	1	1	m_9
$A\bar{B}C\bar{D}$ =	1	0	1	0	1	m_{10}
	1	0	1	1	0	m_{11}
$AB\bar{C}\bar{D}$ =	1	1	0	0	1	m_{12}
	1	1	0	1	0	m_{13}
	1	1	1	0	0	m_{14}
$ABCD$ =	1	1	1	1	1	m_{15}

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3) **K-maps:**

AB \ CD	00	01	11	10
00	m_0 1	m_1 0	m_3 1	m_2 0
01	m_4 0	m_5 1	m_7 0	m_6 1
11	m_{12} 1	m_{13} 0	m_{15} 1	m_{14} 0
10	m_8 0	m_9 1	m_{11} 0	m_{10} 1

→ "As we can see here clearly that there are no possible adjacent groups appearing, and all the 1's are isolated.

→ So, "The K-map gave us no further simplification, so the simplified SOP will be the same as our canonical one".